

Solutions of problems on CH 30

3. Calculate the magnitude of the magnetic field at a point 25.0 cm from a long, thin conductor carrying a current of 2.00 A.

$$B = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(2.00 \text{ A})}{2\pi(0.250 \text{ m})} = \boxed{1.60 \times 10^{-6} \text{ T}}$$

7. A conductor consists of a circular loop of radius $R = 15.0$ cm and two long, straight sections as shown in Figure P30.7. The wire lies in the plane of the paper and carries a current $I = 1.00$ A. Find the magnetic field at the center of the loop.

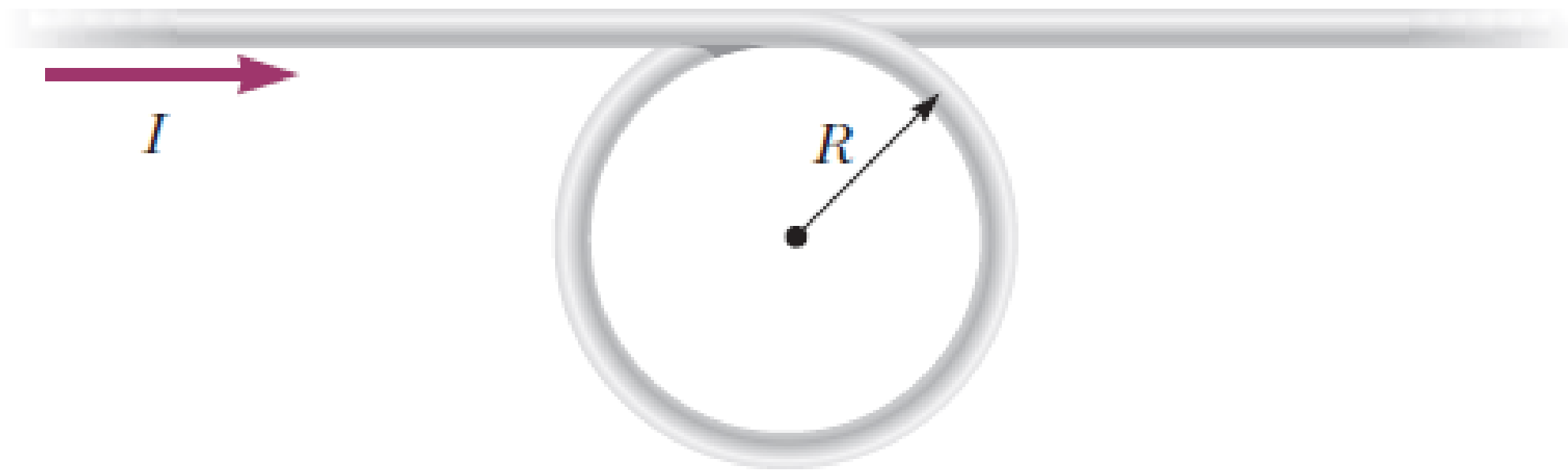


Figure P30.7 Problems 7 and 8.

$$\begin{aligned}\vec{\mathbf{B}} &= \left(1 + \frac{1}{\pi}\right) \frac{\mu_0 I}{2R} = \left(1 + \frac{1}{\pi}\right) \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(1.00 \text{ A})}{2(0.150 \text{ m})} \\ &= 5.52 \times 10^{-6} = \boxed{5.52 \text{ } \mu\text{T} \text{ into the page}}\end{aligned}$$

19. The two wires shown in Figure P30.19 are separated by $d = 10.0$ cm and carry currents of $I = 5.00$ A in opposite directions. Find the magnitude and direction of the net magnetic field (a) at a point midway between the wires; (b) at point P_1 , 10.0 cm to the right of the wire on the right; and (c) at point P_2 , $2d = 20.0$ cm to the left of the wire on the left.

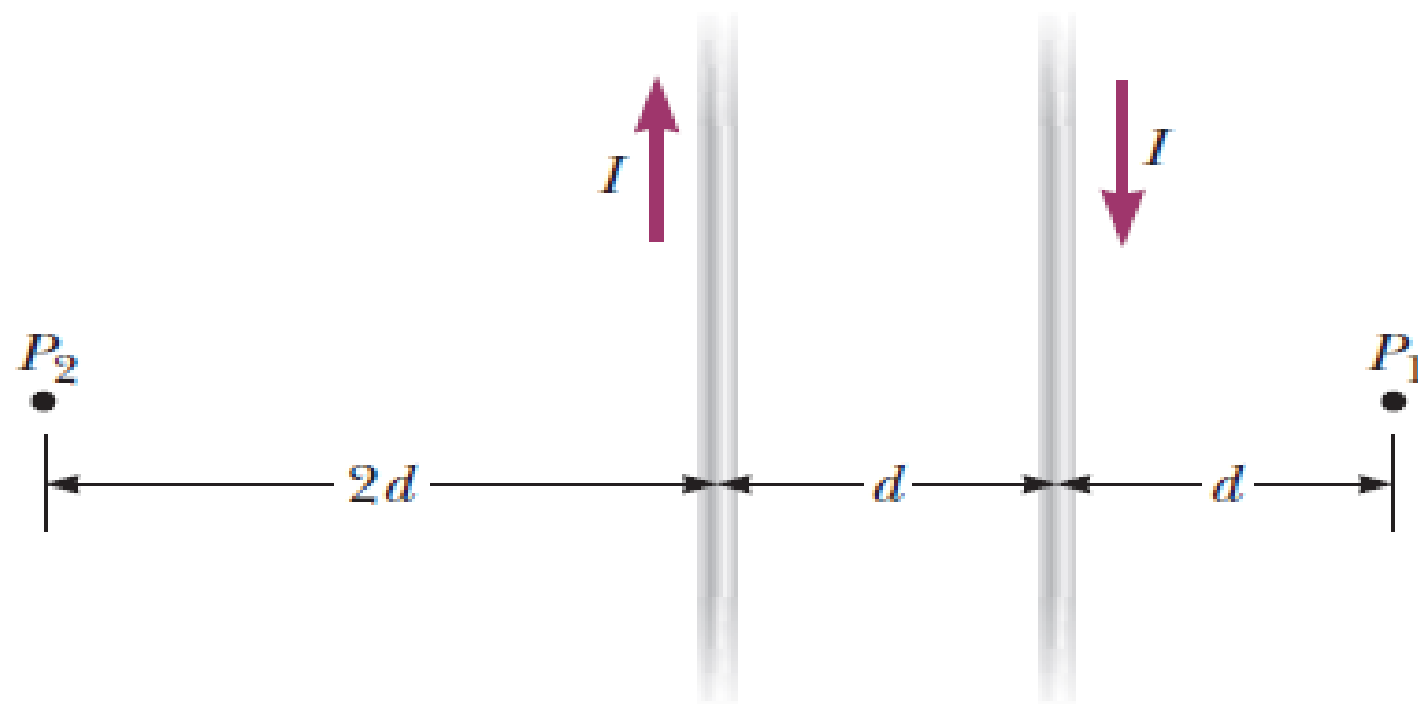


Figure P30.19

(a) At the point half way between the two wires,

$$\begin{aligned} B_{\text{net}} &= -B_1 - B_2 = -\left[\frac{\mu_0 I_1}{2\pi r_1} + \frac{\mu_0 I_2}{2\pi r_2} \right] = -\frac{\mu_0}{2\pi r} (I_1 + I_2) \\ &= -\frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})}{2\pi (5.00 \times 10^{-2} \text{ m})} (10.0 \text{ A}) = -4.00 \times 10^{-5} \text{ T} \end{aligned}$$

or $B_{\text{net}} = \boxed{40.0 \text{ } \mu\text{T} \text{ into the page}}$

(b) At point P_1 ,

$$\begin{aligned} B_{\text{net}} &= +B_1 - B_2 = \frac{\mu_0}{2\pi} \left[+\frac{I_1}{r_1} - \frac{I_2}{r_2} \right] \\ B_{\text{net}} &= \frac{4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}}{2\pi} \left[\frac{5.00 \text{ A}}{0.100 \text{ m}} - \frac{5.00 \text{ A}}{0.200 \text{ m}} \right] \\ &= \boxed{5.00 \text{ } \mu\text{T} \text{ out of page}} \end{aligned}$$

(c) At point P_2 ,

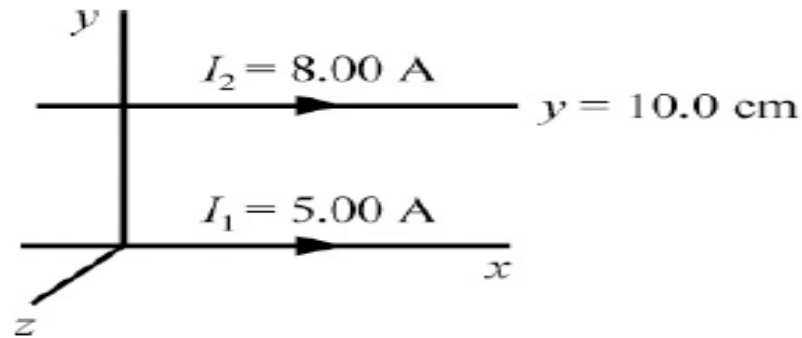
$$B_{\text{net}} = -B_1 + B_2 = \frac{\mu_0}{2\pi} \left[-\frac{I_1}{r_1} + \frac{I_2}{r_2} \right]$$

$$\begin{aligned} B_{\text{net}} &= \frac{4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}}{2\pi} \left[-\frac{5.00 \text{ A}}{0.300 \text{ m}} + \frac{5.00 \text{ A}}{0.200 \text{ m}} \right] \\ &= \boxed{1.67 \text{ } \mu\text{T out of page}} \end{aligned}$$

21. Two long, parallel conductors, separated by 10.0 cm, carry currents in the same direction. The first wire carries a current $I_1 = 5.00$ A, and the second carries $I_2 = 8.00$ A. (a) What is the magnitude of the magnetic field created by I_1 at the location of I_2 ? (b) What is the force per unit length exerted by I_1 on I_2 ? (c) What is the magnitude of the magnetic field created by I_2 at the location of I_1 ? (d) What is the force per length exerted by I_2 on I_1 ?

$$(a) \quad \vec{B} = \frac{\mu_0 I}{2\pi r} \hat{k} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(5.00 \text{ A})}{2\pi(0.100 \text{ m})} \hat{k}$$

$$\vec{B} = \boxed{1.00 \times 10^{-5} \text{ T out of the page}}$$



ANS. FIG. P30.21(a)

$$(b) \quad \vec{F}_B = I_2 \vec{\ell} \times \vec{B} = (8.00 \text{ A}) \left[(1.00 \text{ m}) \hat{i} \times (1.00 \times 10^{-5} \text{ T}) \hat{k} \right] \\ = (8.00 \times 10^{-5} \text{ N}) (-\hat{j})$$



ANS. FIG. P30.21(b)

$$\vec{F}_B = \boxed{8.00 \times 10^{-5} \text{ N toward the first wire}}$$

$$(c) \quad \vec{B} = \frac{\mu_0 I}{2\pi r} (-\hat{k}) = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(8.00 \text{ A})}{2\pi(0.100 \text{ m})} (-\hat{k})$$

$$= (1.60 \times 10^{-5} \text{ T})(-\hat{k})$$



ANS. FIG. P30.21(c)

$$\vec{B} = \boxed{1.60 \times 10^{-5} \text{ T into the page}}$$

$$(d) \quad \vec{F}_B = I_1 \vec{\ell} \times \vec{B} = (5.00 \text{ A}) \left[(1.00 \text{ m}) \hat{i} \times (1.60 \times 10^{-5} \text{ T})(-\hat{k}) \right]$$

$$= (8.00 \times 10^{-5} \text{ N})(+\hat{j})$$



ANS. FIG. P30.21(d)

$$\vec{F}_B = \boxed{8.00 \times 10^{-5} \text{ N towards the second wire}}$$

23. Two parallel wires are separated by 6.00 cm, each carrying 3.00 A of current in the same direction. (a) What is the magnitude of the force per unit length between the wires? (b) Is the force attractive or repulsive?

(a) $F/\ell = \mu_0 I_1 I_2 / 2\pi d$

$$\frac{F}{\ell} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(3.00 \text{ A})^2}{2\pi(6.00 \times 10^{-2} \text{ m})} = \boxed{3.00 \times 10^{-5} \text{ N/m}}$$

(b) attractive

32. The magnetic coils of a tokamak fusion reactor are **W** in the shape of a toroid having an inner radius of 0.700 m and an outer radius of 1.30 m. The toroid has 900 turns of large-diameter wire, each of which carries a current of 14.0 kA. Find the magnitude of the magnetic field inside the toroid along (a) the inner radius and (b) the outer radius.



$$(a) \quad B_{\text{inner}} = \frac{\mu_0 NI}{2\pi r} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(900)(14.0 \times 10^3 \text{ A})}{2\pi(0.700 \text{ m})} = \boxed{3.60 \text{ T}}$$

$$(b) \quad B_{\text{outer}} = \frac{\mu_0 NI}{2\pi r} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(900)(14.0 \times 10^3 \text{ A})}{2\pi(1.30 \text{ m})} = \boxed{1.94 \text{ T}}$$

- 47.** A cube of edge length $\ell = 2.50$ cm is positioned as shown in Figure P30.47. A uniform magnetic field given by $\vec{\mathbf{B}} = (5\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 3\hat{\mathbf{k}})$ T exists throughout the region. (a) Calculate the magnetic flux through the shaded face. (b) What is the total flux through the six faces?

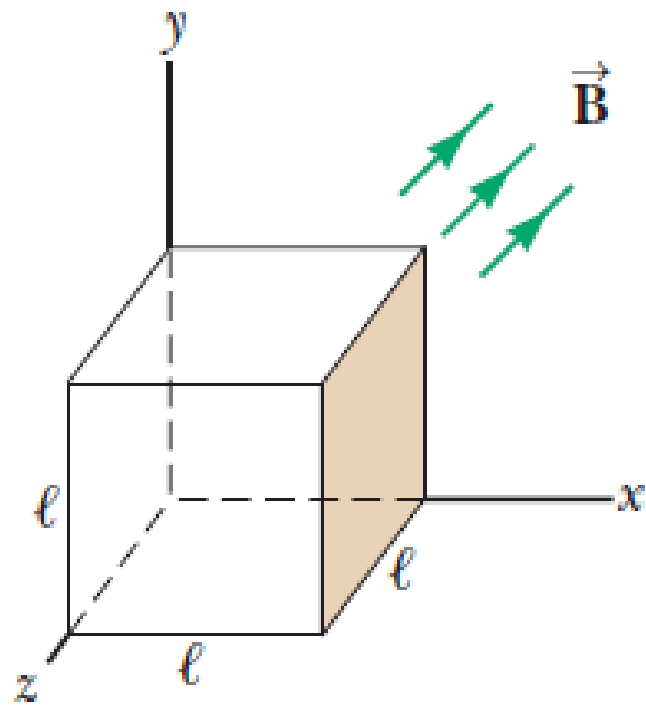


Figure P30.47

The flux is defined as $\Phi_B = \vec{B} \cdot \vec{A}$

- (a) The flux through the shaded face is $\Phi_B = B_x A_x + B_y A_y + B_z A_z$. The shaded square's area is in the yz plane, so it counts as an x component of area. Here $A_y = A_z = 0$. Then,

$$\Phi_B = \int \vec{B} \cdot d\vec{A} = \vec{B} \cdot \vec{A} = (5\hat{i} + 4\hat{j} + 3\hat{k}) \text{ T} \cdot (2.50 \times 10^{-2} \text{ m})^2 \hat{i}$$

$$\Phi_B = 3.12 \times 10^{-3} \text{ T} \cdot \text{m}^2 = 3.12 \times 10^{-3} \text{ Wb} = \boxed{3.12 \text{ mWb}}$$

- (b) For a closed surface, $\oint \vec{B} \cdot d\vec{A} = 0$, so $(\Phi_B)_{\text{total}} = \oint \vec{B} \cdot d\vec{A} = \boxed{0}$

51. A 30.0-turn solenoid of length 6.00 cm produces a magnetic field of magnitude 2.00 mT at its center. Find the current in the solenoid.

$$I = \frac{BL}{\mu_0 N} = \frac{(2.00 \times 10^{-3} \text{ T})(6.00 \times 10^{-2} \text{ m})}{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(30.0)} = \boxed{3.18 \text{ A}}$$